# J-GATE3.0
# ITCH Connectivity Manual

Version 2.1

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Osaka Exchange, INC

Tokyo Commodity Exchange, INC

While the utmost care has been taken to ensure the accuracy of the details of this connectivity manual (hereinafter referred to as “Manual”), we shall not be liable to any persons or entities for any disadvantages or problems whatsoever arising from the use of information contained in the Manual.

The details of the Manual may be subject to change. The modified points are released as the modified version in a timely manner. When developing the system connected to J-GATE, the latest version should be used for Manual and the relevant document.

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# 1. About this Manual

## 1.1 Introduction

This manual is an interface specification manual for the high-frequency market information provided by the J-GATE3.0 system (hereinafter referred to as "J-GATE"). It provides information needed primarily by the following groups: trading participants at Osaka Exchange, Inc. (hereinafter referred to as "the Osaka Exchange") and at the Tokyo Commodity Exchange, Inc. (hereinafter referred to as “TOCOM”), as well as information vendors and software vendors (hereinafter referred to as "users") who develop applications used to acquire high-frequency market information.

### 1.2 Overview of the messages provided by the system

The J-GATE system uses the ITCH protocol to provide high-frequency market information. ITCH disseminates data on all orders and trades that are published on the J-GATE. ITCH primarily provides the following types of data:

・Order book information: Data on all orders is provided in the standard ITCH format. Although the data provided includes execution information, information on J-NET trades is not disseminated.

・Series information: Series information is provided, such as Series ID and 9-digit codes. Data related to tick size is also provided.

・Session information: Session information is provided, such as order acceptance start, morning session start, and trading halt.

We make every effort to prevent an incident in J-GATE. However, even if ITCH is down, we, in principle, continue trading as long as OMnet API is running to provide market data. (Except in cases where more than a certain ratio of trading participants cannot participate in trading.)

## 1.3 Protocol used for the provided messages

An ITCH message consists of a series of sequenced messages, each of which has a different length based on its type. ITCH messages are binary encoded using SoupBinTCP or MoldUDP64.

ITCH is a protocol developed to guarantee message sequencing and delivery.

| Protocol type | Explanation |
|---|---|
| SoupBinTCP | This protocol is used to send and receive snapshots.<br>SoupBinTCP allows delivery of a set of sequenced messages from a server to a client in real time. Because SoupBinTCP assigns sequence numbers to messages, it guarantees message sequencing even if the connection to the target TCP/IP socket fails.<br>The sequence numbers are implicit, and the client needs to maintain a counter that is increased each time a message is received. At reconnection after a connection failure, the client submits the last-seen sequence number in its login message, and the server resends every message starting from that sequence number. |
| MoldUDP64 | This protocol is used to receive ITCH messages and send resend requests.<br>MoldUDP64 is a network protocol that is higher than UDP and provides a mechanism for listeners to detect and re-request missed packets. Unlike SoupBinTCP, MoldUDP64 assigns an explicit sequence number to each message. If a packet loss is detected by the client, it can re-request that packet from the MoldUDP64 gateway, and the packet is then resent as a UDP unicast to that client. |

## 1.4 Handling of this manual on Holiday Trading day

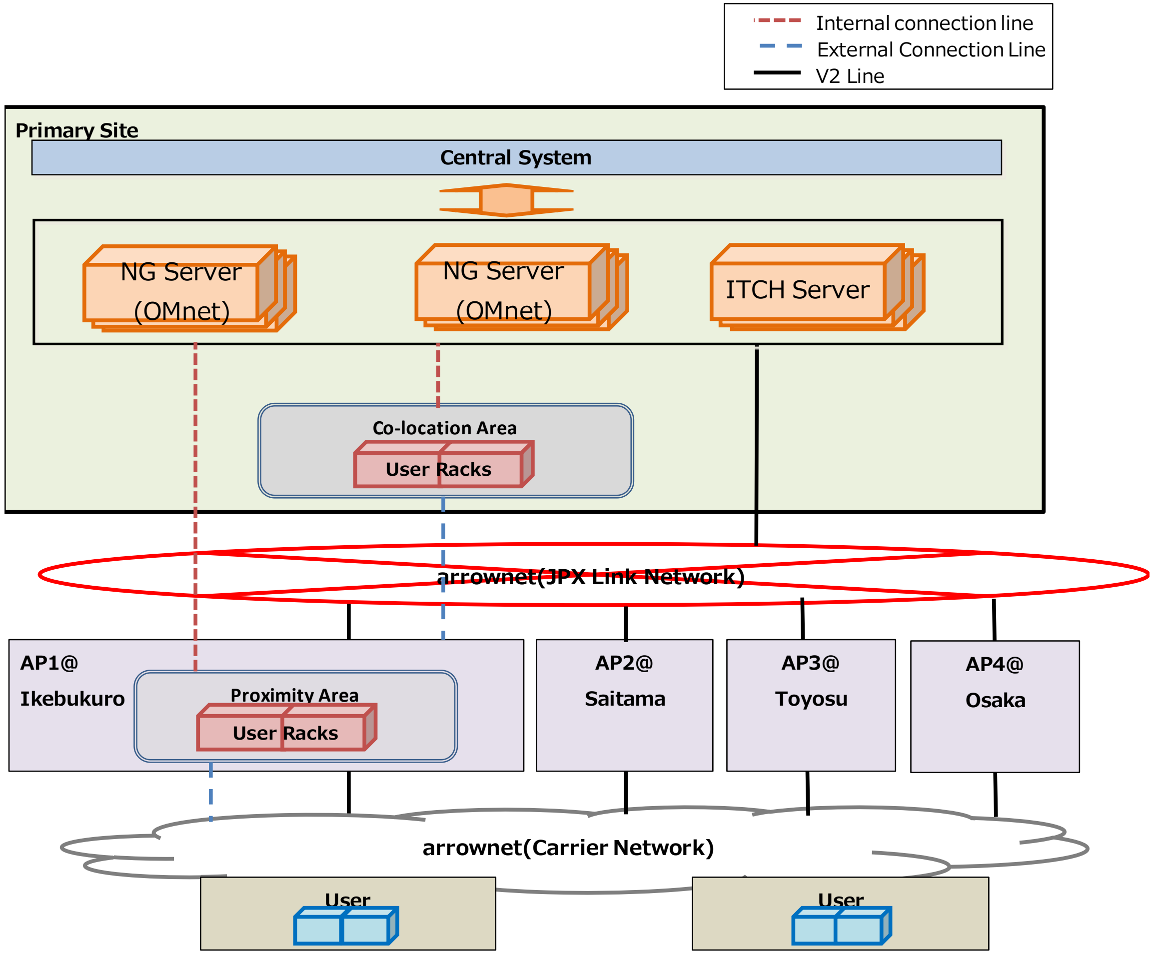
What’s written in this manual also applies to Holiday Trading days that will be introduced from September 2022.

# 2. Connection Method

## 2.1 Overall configuration

In the J-GATE, an ITCH user connects to the ITCH gateway (ITCH NG server) via arrownet (a wide area network that connects the Exchange to individual user sites), and then communicates with the central system. The following figure provides an overview.

Image_2.1 J-GATE Overview



## 2.2 ITCH server configuration

The ITCH server allocates products to be processed to multiple servers, which distribute data on the target products in parallel. In addition, a redundant configuration is used in which two servers distribute the same data on the same product.

Furthermore, three data transmission processes (send, rewind, and glimpse) are started on each ITCH server. Each process sends multicast data, resend data, and snapshot data.

The following figure shows the ITCH server configuration. The table below it provides an overview of the transmitted data.

Image_2.2 ITCH Server Example (Nikkei 225 Options, Nikkei 225 mini Options)

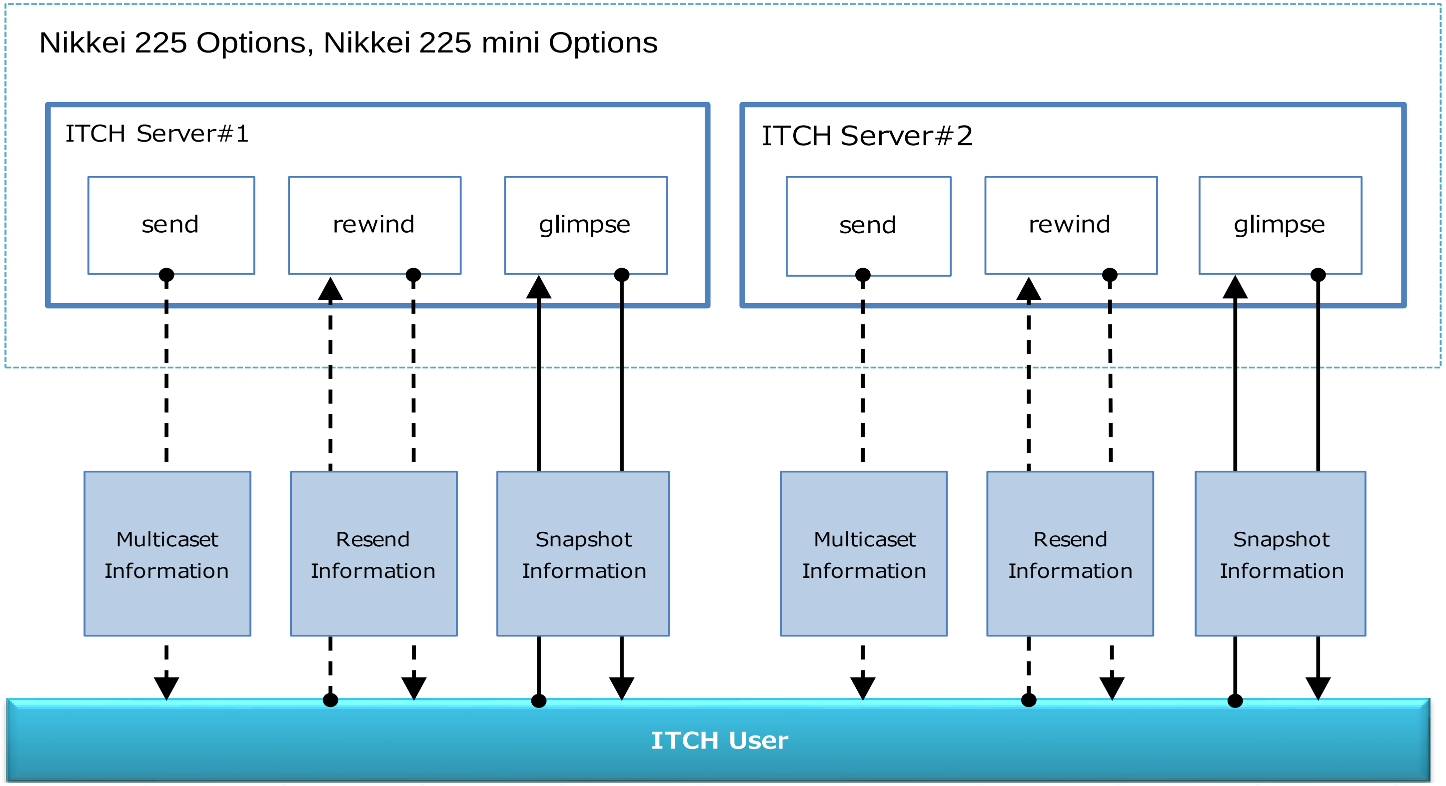


Table_2.2 Overview of the transmitted data

| No. | Type of transmitted data | Protocol used | | ITCH server's dissemination process | Overview of the transmitted data | Remarks |
|---|---|---|---|---|---|---|
| 1 | Multicast data | UDP multicast | MoldUDP64 | send | Data such as order book information and execution information that is generated in the J-GATE and broadcast to all ITCH users whenever data is generated. | |
| 2 | Resend data | UDP unicast | MoldUDP64 | rewind | Using a resend request from an ITCH user as the trigger, the ITCH server resends data to the requesting ITCH user.<br>(This option is used when the ITCH user detects that the aforementioned multicast data has been lost.) | |
| 3 | Snapshot data | TCP unicast | SoupBinTCP | glimpse | Using a snapshot request from an ITCH user as the trigger, the ITCH server sends to the requesting user the order book information and other data that is the most recent at the time of the request.<br>(If an extended failure occurs on the ITCH user side, this option is used to obtain only the most recent data after recovery, rather than receiving all data up to that point in time.) | |

# 3. Communication Method

## 3.1 Connection conditions

Table_3.1 Connection conditions

| No. | Item | Description | Remarks |
|---|---|---|---|
| 1 | DSU/ONU | Per carrier specifications | |
| 2 | Transmission method | Transparent mode | |
| 3 | Communication method | Full duplex transmission | |
| 4 | Transmission control protocol | UDP (MoldUDP64) and TCP (SoupBinTCP) | For the specifications of MoldUDP64 and SoupBinTCP, refer to each manual. |
| 5 | Transmission code | Single-byte alphanumeric: ASCII code | |
| 6 | Receiving-side router | IP multicast support<br>IGMP version: IGMPv2<br>Multicast routing protocol: PIM-SM | |

* IGMP Version3 message may be sent to the receiving-side router to confirm if it's been used or not, but it can be discarded as it is not necessary for operation.

# 4. Communication Specifications
## 4.1 Common specifications

This chapter describes the communication specifications that are common to multicast data, resend data, and snapshot data.

(1) Packet structure

The following figure shows the packet structure on the transmission route.

| Ethernet header | IP header | UDP/TCP header | User packet |
|---|---|---|---|

The following subsections (4.2 through 4.4) describe the setting details for the transmission packet, which is referred to as the "User packet" in the figure above.

(2) Data format

The following table shows the data formats that are set in the user packet.

Table_4.1 Data format

| No. | Data format | Size | Explanation | Remarks |
|---|---|---|---|---|
| 1 | Alpha | Variable | Set left-aligned and filled to the right with spaces | |
| 2 | Numeric | 1, 2, 4 or 8 bytes | Signed integer encoded in big-endian binary | |
| 3 | Price | 4 bytes | Signed integer encoded in big-endian binary | |
| 4 | Date | 4 bytes | Signed integer encoded in big-endian binary which is set in YYYYMMDD format. | |

## 4.2 Communication specifications related to multicast data

### 4.2.1 Overview

Multicast data is disseminated using UDP multicast, with a different multicast group number assigned to each product group. An ITCH user participates in the multicast group for the desired product and acquires order book information and execution information for that product.

Furthermore, two ITCH servers output the same data at the same time to lines on two systems (systems 1 and 2). Therefore, even if a failure occurs in system 1 or its server, output continues on system 2, allowing the ITCH user to maintain data continuity.

### 4.2.2 Maximum number of messages per second and maximum bandwidth for multicast groups

The assumed maximum value of the number of ITCH distributions related to new orders and cancellation orders in Zaraba, and the calculation of the assumed required line bandwidth is shown. In addition, the series information tags that are distributed when joining a multicast group at the beginning of the day and session transition tags that are distributed at the time of session transition are distributed in large quantities instantaneously are described for reference.

Note that the numerical values shown are assumed values and do not guarantee the upper limit of the actual flow rate. It may momentarily exceed the assumed values depending on market conditions. If disseminated messages exceed the assumed bandwidth and a packet loss is detected, send a rewind request to recover the missed information.

- Calculating the maximum number of ITCH distributions in Zaraba and the assumed required bandwidth

  The estimated maximum number of contracts per second for new orders and cancellation orders distributed by ITCH is calculated for each partition based on the maximum number of orders per second in J-GATE. The required bandwidth is calculated by taking into account the message length of the new order tag (A-tag) and the cancel order tag (D-tag).

Table_4.2.2-1 Assumed ITCH maximum number per second and the assumed required bandwidth

<table>
<tr><th>P</th><th>MCG</th><th>Product</th><th>Max No. Orders/sec (overall)</th><th>Individual</th><th>Max No. Distributions/sec (Overall)</th><th>Individual</th><th>Message Length (byte) (*1)</th><th>Line Bandwidth (Mbps) (Overall)</th><th>Individual</th></tr>
<tr><td>1</td><td>1</td><td>Nikkei 225 Options, Nikkei 225 mini Options</td><td rowspan="8">105,314</td><td>28,282</td><td rowspan="8">314,898</td><td>95,536</td><td>28</td><td rowspan="8">69.3</td><td>21.0</td></tr>
<tr><td>2</td><td>2</td><td>Nikkei 225 mini</td><td>28,282</td><td>95,536</td><td>28</td><td>21.0</td></tr>
<tr><td rowspan="2">3</td><td>3</td><td>Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei400 Futures</td><td rowspan="2">28,282</td><td rowspan="2">95,536</td><td rowspan="2">28</td><td rowspan="2">21.0</td></tr>
<tr><td>4</td><td>Other Index Futures/Options</td></tr>
<tr><td rowspan="3">4</td><td>5</td><td>Single Stock Options</td><td rowspan="3">28,282</td><td rowspan="3">95,536</td><td rowspan="3">28</td><td rowspan="3">21.0</td></tr>
<tr><td>6</td><td>JGB Futures, JGB Futures Options, Interest Rate Futures</td></tr>
<tr><td>7</td><td>OSE Commodity Futures (Precious Metals, Agricultural, Rubber, Gold Futures Options)</td></tr>
<tr><td>5</td><td>8</td><td>TOCOM Commodity Futures (Crude Oil, Power, LNG)</td><td>28,282</td><td>95,536</td><td>28</td><td>21.0</td></tr>
</table>

（*1）The value of the A-Tag added to the D-Tag multiplied by 1/2.

➢ (Reference） Calculation of assumed required line bandwidth when series information R tags etc. are distributed instantaneously

Estimate the number of messages that can be generated based on the number of tradeable series, the assumed execution rate (BD70 and BD2) at the time of auction, and the percentage of a series' remaining orders after auction (BO16 / BO17). In addition to this, the required line bandwidth is calculated in consideration of the message length of each message.

Table_4.2.2-2 Assumed required bandwidth when series information R tag and such is disseminated

<table>
<tr><th rowspan="2">P</th><th rowspan="2">MCG</th><th rowspan="2">Product</th><th rowspan="2">No. of Series (∗)</th><th colspan="3">Message Length (byte)</th><th rowspan="2">Line Bandwidth (Mbps) (Overall</th><th rowspan="2">Individual</th></tr>
<tr><th>R Tag</th><th>L Tag</th><th>O Tag</th></tr>
<tr><td>1</td><td>1</td><td>Nikkei 225 Options, Nikkei 225 mini Options</td><td>2,996</td><td>129</td><td>25</td><td>29</td><td rowspan="8">28.1</td><td>4.4</td></tr>
<tr><td>2</td><td>2</td><td>Nikkei 225 mini</td><td>16</td><td>129</td><td>25</td><td>29</td><td>0.0</td></tr>
<tr><td rowspan="2">3</td><td>3</td><td>Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei400 Futures</td><td rowspan="2">1,849</td><td rowspan="2">129</td><td rowspan="2">25</td><td rowspan="2">29</td><td rowspan="2">2.7</td></tr>
<tr><td>4</td><td>Other Index Futures/Options</td></tr>
<tr><td rowspan="3">4</td><td>5</td><td>Single Stock Options</td><td rowspan="3">14,148</td><td rowspan="3">129</td><td rowspan="3">25</td><td rowspan="3">29</td><td rowspan="3">20.7</td></tr>
<tr><td>6</td><td>JGB Futures, JGB Futures Options, Interest Rate Futures</td></tr>
<tr><td>7</td><td>OSE Commodity Futures (Precious Metals, Agricultural, Rubber, Gold Futures Options)</td></tr>
<tr><td>5</td><td>8</td><td>TOCOM Commodity Futures (Crude Oil, Power, LNG)</td><td>174</td><td>129</td><td>25</td><td>29</td><td>0.3</td></tr>
</table>

(*) The number of tradable series in the production environment at some point in the past.

Note 1: Business Header (20byte) is not considered

Note 2: The above calculation does not take into account the message length of the API header. Further, it is preferable that the line bandwidth secured by the user, considers line efficiency, etc.

### 4.2.3 IP addresses and port numbers

The following table shows the IP addresses and port numbers.

(1) For production

Table_4.2.3-1 IP addresses and port numbers

<table>
<tr><th>P</th><th>Multicast group</th><th>Product</th><th>IP address (System 1)</th><th>IP address (System 2)</th><th>Transmission destination port number (System 1)</th><th>Transmission destination port number (System 2)</th></tr>
<tr><td>1</td><td>MCG001</td><td>Nikkei 225 Options, Nikkei 225 mini Options</td><td>239.194.41.1</td><td>239.194.42.1</td><td>21001</td><td>21101</td></tr>
<tr><td>2</td><td>MCG002</td><td>Nikkei 225 mini</td><td>239.194.41.2</td><td>239.194.42.2</td><td>21002</td><td>21102</td></tr>
<tr><td rowspan="2">3</td><td>MCG003</td><td>Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei400 Futures</td><td>239.194.41.3</td><td>239.194.42.3</td><td>21003</td><td>21103</td></tr>
<tr><td>MCG004</td><td>Other Index Futures/Options</td><td>239.194.41.4</td><td>239.194.42.4</td><td>21004</td><td>21104</td></tr>
<tr><td rowspan="3">4</td><td>MCG005</td><td>Single Stock Options</td><td>239.194.41.5</td><td>239.194.42.5</td><td>21005</td><td>21105</td></tr>
<tr><td>MCG006</td><td>JGB Futures, JGB Options, Interest Rate Futures</td><td>239.194.41.6</td><td>239.194.42.6</td><td>21006</td><td>21106</td></tr>
<tr><td>MCG007</td><td>OSE Commodity Futures (Precious Metals, Agricultural, Rubber, Gold Futures Options)</td><td>239.194.41.7</td><td>239.194.42.7</td><td>21007</td><td>21107</td></tr>
<tr><td>5</td><td>MCG008</td><td>TOCOM Commodity Futures (Crude Oil, Power, LNG)</td><td>239.194.41.8</td><td>239.194.42.8</td><td>21008</td><td>21108</td></tr>
</table>

Table_4.2.3-2 Multicast Rendezvous Point (RP)

| System | Multicast RP Address |
|---|---|
| System 1 | 10.19.3.41 |
| System 2 | 10.19.3.42 |

(2) For production after a disaster recovery

[1] Same as production

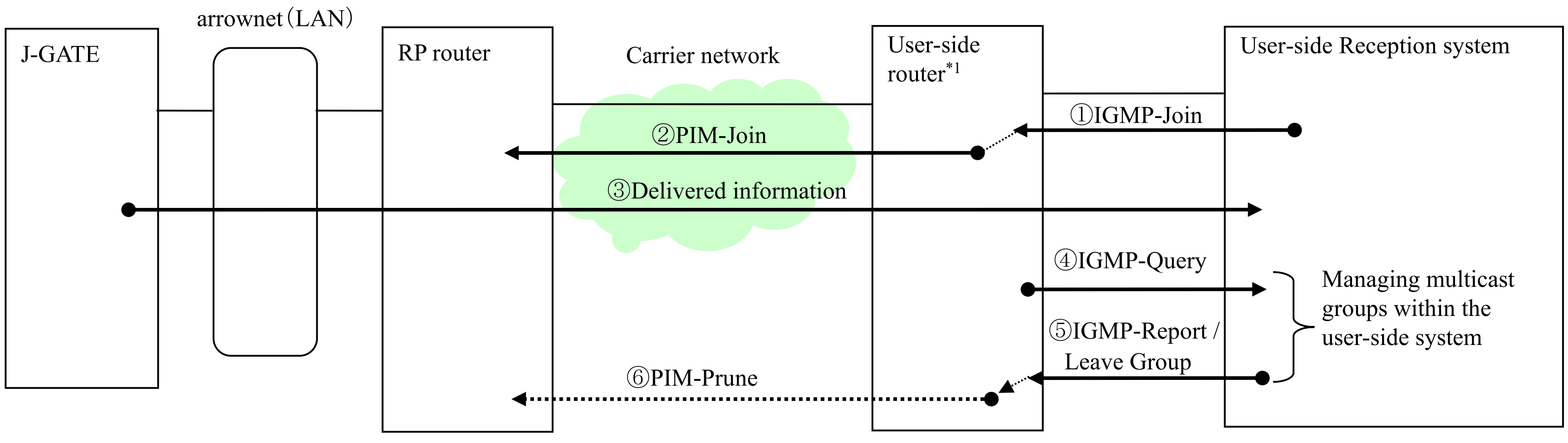
(3) For testing on market holidays

[1] Same as production

### 4.2.4 Participation in and withdrawal from a multicast group

To receive data disseminated by the ITCH server, an ITCH user must participate in a multicast group for the applicable product. The following figure shows the procedure for participating in and withdrawing from a multicast group:

Image_4.2.4　Participation in and withdrawal from a multicast group




①User-side Reception system participates in a multicast group (sends IGMP-Join to the user-side router).
②User-side router declares participation in a multicast tree (sends PIM-Join to the RP router).
③J-GATE disseminates information to the multicast group.
④User-side router asks the user-side reception system if it is joining in a multicast group (sends IGMP-Query).
⑤User-side reception system reports the status of multicast group participation (sends IGMP-Report or Leave Group).
⑥When user-side router detect that there is no node participating in a multicast group, it declares withdrawal from that multicast group (sends PIM-Prune to the RP router).

*1: IP address of RP router for the multicast group to participate in must be set in the user-side router.

### 4.2.5 Content of the packets that are transmitted

(1) Types of packets that are transmitted

The following table shows the types of packets that are transmitted.

Table_4.2.5-1 Types of packets that are transmitted

| No. | Packet type | Transmission direction | Packet content | Transmission timing | Remarks |
|---|---|---|---|---|---|
| 1 | Message packet | ITCH server→ ITCH user | Stores one or more message blocks. | Varies depending on the content of the messages stored in the packet.<br>See the timing for tag output in *6. Data Format*. | |
| 2 | Heartbeat packet | ITCH server→ ITCH user | Used by an ITCH user to verify that communication from the ITCH server to the ITCH user is continuous. | Sent in multicast group units between 06:24 am on a business day and 06:20 am on the following day.<br>Sent when a message packet has not been sent for 0.05 seconds. | |

(2) Settings

(a) Message packet

A message packet consists of a packet header and one or more message blocks.

Table_4.2.5-2　Settings

| No. | Item name | | | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|---|---|
| 1 | Packet header | Session | | 0 | 10 | Alpha | The session data is set.<br>If a system reboot happens during the day, the session changes, and the session before the reboot can no longer be used.<br>Set in the format of “JPxxxxxxxx”<br>JP　　: Fixed value<br>xxxxxxxx: Linux time (in 8-digit numeral) which the system sets when the online operation starts. The value is fixed until the end of online operation.<br><br>e.g., JP23839531 | |
| 2 | | Sequence Number | | 10 | 8 | Numeric | The sequence number of the first message block in the packet is set.<br>This sequence number begins with 1 within each multicast group unit and is reset to 1 when the session changes. | |
| 3 | | Message Count | | 18 | 2 | Numeric | The number of message blocks stored inside the packet is set. | |
| 4 | Message block (1) | Data message header | Message Length | Variable | Numeric | Numeric | The length of message block (1) (excluding the Message Length field) is set in bytes. | |
| 5 | | Message header | Message Data | Variable | - | - | The message data (application data that begins with “Message Type”) is set.<br>For details about the content, see *6. Data Format*. | |
| 6 | ・・・ | | | | | | | |
| 7 | Message block (*n*) | Data message header | Message Length | Variable | 2 | Numeric | Same as message block (1). | |
| 8 | | Message header | Message Data | Variable | - | - | | |

(b) Heartbeat packet

Table_4.2.5-3 Heartbeat packet

| No. | Item name | | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Packet header | Session | 0 | 10 | Alpha | Same as *Session* in the packet header. | |
| 2 | | Sequence Number | 10 | 8 | Numeric | The sequence number of the first message block in the next message packet is set.<br>Since the sequence number is incremented according to the number of message blocks, it is not incremented when a heartbeat packet is sent. | |
| 3 | | Message Count | 18 | 2 | Numeric | “0” is set. | |

### 4.2.6 Other

(1) Since UDP multicast is a protocol that does not guarantee packet arrival, the ITCH user must take the following steps:

[1] If no message packet or heartbeat packet is received for a certain time, assume that some failure has occurred and switch to the other system to receive data.

[2] Use the sequence number stored in the packet header of the message packet to check for packet continuity. If packet loss is detected, use the method described in *4.3 Communication specifications related to resend data* to send a resend request to the ITCH server.

(2) The message packet stores multiple message blocks in a single Ethernet frame, but the number of message blocks in System 1 and System 2 may not always be the same. For example, in System 1, Message blocks (1) through (10) may be stored in a single Ethernet frame, while in System 2, Message blocks (1) through (8) may be stored in one Ethernet frame and Message blocks (9) and (10) may be stored in a second Ethernet frame.

However, the sequence number in the packet header of the message packet will be the same, guaranteeing packet continuity.

(3) The multicast data (excluding Heartbeat packets) will be delivered between 06:24 (*) on a business day and 06:20 on the following day; be sure to have the join request message for multicast group sent from the ITCH user-side router to the RP, and be able to receive the multicast data before 06:24 am on a business day.

*Please note that time may differ due to Night Batch processing.

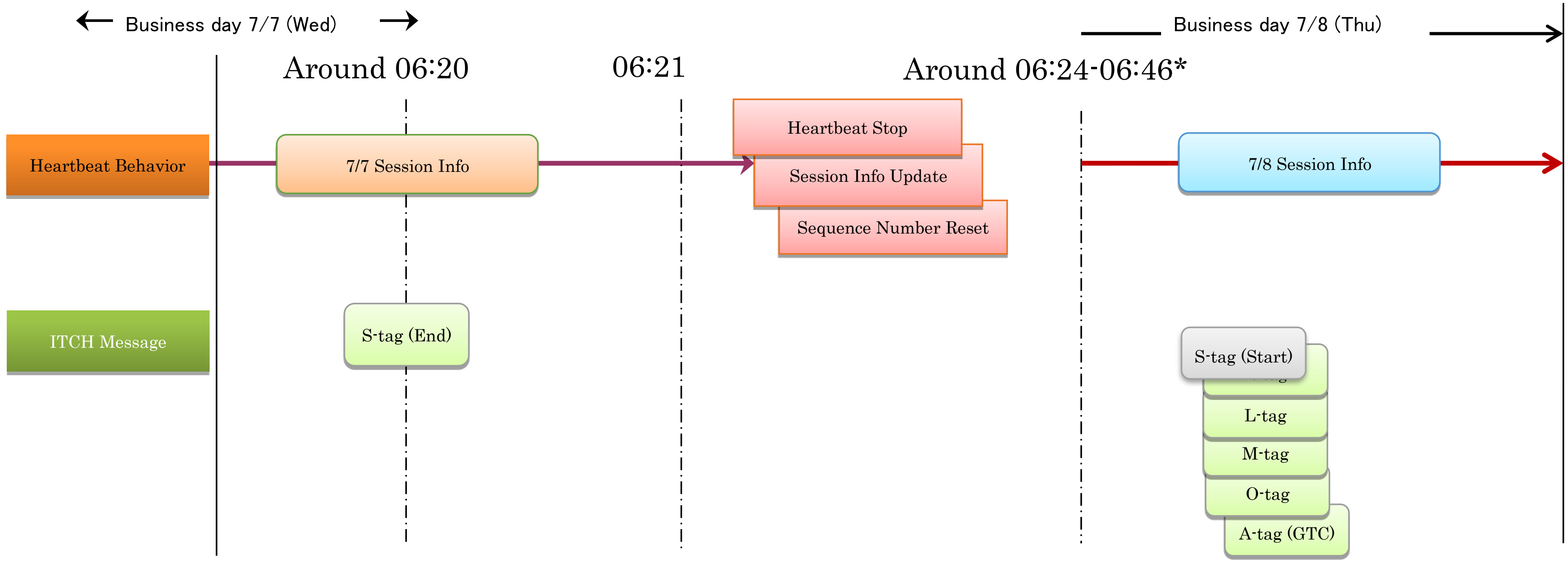
(4) Heartbeat messages are sent continuously after Event Code C is set to S-tag.

Heartbeat stops when J-GATE Night Batch starts (around 06:20)*

When Heartbeat restarts, Session information will be updated with the next day’s information, and Sequence Number will reset (around 06:24-06:46)*.

After sending the Start Message with Event Code 0 set to S-tag, Series information, Order price information, Leg information, and Session information will be sent, and Reload information of GTC Orders will be distributed.

Image_4.2.6　Heartbeat message




*Please note that time may differ due to Night Batch processing. If the planned online start time becomes later than7:15, OSE will notify users.

*If hardware replacement is needed due to J-GATE hardware failure and such, the Night Batch start time might get delayed. If there is any hardware replacement, users will be notified in advance.

## 4.3. Communication specifications related to resend data

### 4.3.1 Overview

If the ITCH user detects the loss of a UDP multicast packet as described in *4.2 Communication specifications related to multicast data* (that is, when the continuity of sequence numbers has been disrupted), the ITCH user can receive the lost packet by sending a resend request to the ITCH server. Communication between the ITCH server and the ITCH user uses UDP unicast.

### 4.3.2 IP addresses and port numbers

The following table shows the IP addresses and port numbers:

(1) For production operation

Table_4.3.2 IP addresses and port numbers

| System | Multicast Group | Product | ITCH Server-side IP address | ITCH Server-side Port number |
|---|---|---|---|---|
| System 1 | MCG001 | Nikkei 225 Options, Nikkei 225 mini Options | 10.18.18.1 | 24002 |
| | MCG002 | Nikkei 225 mini | 10.18.18.3 | 24003 |
| | MCG003 | Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei 400 Futures | 10.18.18.5 | 24004 |
| | MCG004 | Other Index Futures/Options | 10.18.18.5 | 24108 |
| | MCG005 | Single Stock Options | 10.18.18.7 | 24005 |
| | MCG006 | JGB Futures, JGB Futures Options, Interest Rate Futures | 10.18.18.7 | 24110 |
| | MCG007 | OSE Commodity Futures (Precious Metals, Agriculture, Rubber, Gold Futures Options, etc.) | 10.18.18.7 | 24215 |
| | MCG008 | TOCOM Commodity Futures (Crude Oil, Power, LNG) | 10.18.18.9 | 24006 |
| System 2 | MCG001 | Nikkei 225 Options, Nikkei 225 mini Options | 10.18.19.2 | 24002 |
| | MCG002 | Nikkei 225 mini | 10.18.19.4 | 24003 |
| | MCG003 | Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei400 Futures | 10.18.19.6 | 24004 |
| | MCG004 | Other Index Futures/Options | 10.18.19.6 | 24108 |
| | MCG005 | Single Stock Options | 10.18.19.8 | 24005 |
| | MCG006 | JGB Futures, JGB Futures Options, Interest Rate Futures | 10.18.19.8 | 24110 |
| | MCG007 | OSE Commodity Futures (Precious Metals, Agriculture, Rubber, Gold Futures Options, etc.) | 10.18.19.8 | 24215 |
| | MCG008 | TOCOM Commodity Futures (Crude Oil, Power, LNG) | 10.18.19.10 | 24006 |

(2) For production operation after disaster recovery

Same as (1) For production operation

(3) For testing on market holidays

Same as (1) For production operation

### 4.3.3 Resend procedure

Packets are resent in the following this procedure:

(1) The ITCH user detects a loss of UDP multicast packet.

(2) The ITCH user sends the ITCH server a resend request packet in which *the first sequence number being requested,* and *the number of data messages being requested* are set.

(3) After receiving the resend request packet, the ITCH server sends the ITCH user a message packet (a single packet) that stores the applicable data messages. (If the requested data messages do not fit inside a single packet, not all of them are sent.)

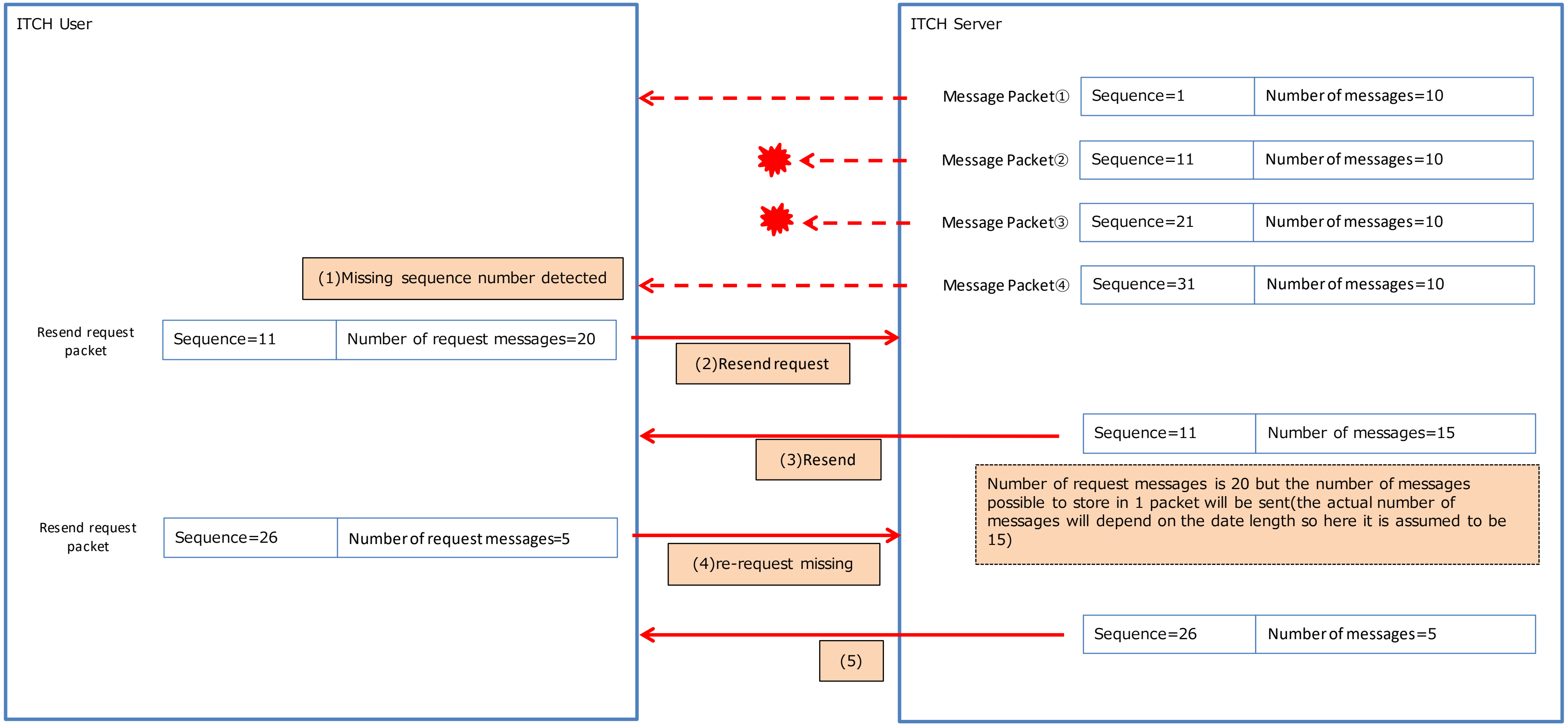
(4) If the data messages in the packet that the ITCH user receives are fewer than were requested, the ITCH user sends the ITCH server a resend request packet in which the succeeding sequence number and number of data messages are set.

(5) The ITCH server sends the applicable message packet (a single packet) to the ITCH user. (Thereafter, these steps are repeated until the ITCH user receives all the data messages it requested.)

If the value of the resend request packet sent by the ITCH user is invalid (for example, the format is invalid or the specified sequence number does not exist), the ITCH server discards that request packet.

The following figure provides an overview of the resend procedure.

Image_4.3.3 Resend process flow overview




The numbers in parentheses correspond to the numbers in the explanation on the previous page.

### 4.3.4 Content of the packets that are transmitted

(1) Types of packets that are transmitted

The following table shows the types of packets that are transmitted.

Table_4.3.4-1 Transmitted packet

| No. | Packet type | Transmission direction | Packet content | Transmission timing | Remarks |
|---|---|---|---|---|---|
| 1 | Resend request packet | ITCH user→ ITCH server | Used to indicate the first sequence number and number of data messages being requested by the ITCH user. | When the ITCH user detects a packet loss | |
| 2 | Resend packet | ITCH server→ ITCH user | Stores the data message that corresponds to the resend request from the ITCH user. | When a resend request packet is received from the ITCH user | |

(2) Settings

(a) Resend request packet

Table_4.3.4-2 Resend request packet

| No. | Item | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Session | 0 | 10 | Alpha | Same as *Session* in the packet header.<br>Only the session number of the current day can be set. | |
| 2 | Sequence Number | 10 | 8 | Numeric | The sequence number of the first data message being requested for resend is set. | |
| 3 | Requested Message Count | 18 | 2 | Numeric | The number of data messages being requested for resend is set. | |

(b) Resend packet

The packet is sent in the same format as the message packet described in *4.2 Communication specifications related to multicast data*.

The resend packet that is sent stores the number of message blocks that can be stored in a single Ethernet frame. Consequently, even if the original data was sent as multicast data in multiple packets (multiple Ethernet frames) the same data might be resent in a single packet (a single Ethernet frame).

#### 4.3.5 Other

(1) Packets can be resent only while the ITCH server is online (that is, between 06:39 on a business day and 06:20 on the following day)).

(2) Only data of the current day from after the login can be obtained by resend request.

(3) As Resend Request function strains the ITCH server, it should only be used for detecting lost packets for the latest UDP Multicast.

In cases where Multicast information cannot be received for a long time due to failure, please use Snapshot's obtain information function mentioned in the next section.

(4) JPX Market Innovation & Research, Inc. has introduced Storm Control functionality to unicast communications in JPX co-location internal connection lines. Per (3), this functionality is supposed to be used only to retrieve lost messages of the most recent UDP multicast data, but please be fully aware that Storm Control functionality may be triggered, if a large number of requests are sent in an extremely short period of time, regardless of the user's intention.

### 4.4. Communication specifications related to snapshot data

#### 4.4.1 Overview

After recovery from a failure that had prevented multicast data from being received over an extended period, the ITCH user may choose to obtain only the most recent data (snapshot data) instead of receiving all the data again. By sending a snapshot request packet to the ITCH server, the ITCH user can receive the latest snapshot data.

To send a snapshot request, the ITCH user must log in to the ITCH server, using the unique user ID and login password assigned to it.

Communication between the ITCH user and the ITCH server uses TCP unicast.

As the snapshot data acquisition process strains the ITCH server, only use it when a participant-side system failure occurs.

### 4.4.2 IP addresses and port numbers

The following table shows the IP addresses and port numbers.

(1) For production operation

Table_4.4.2 IP addresses and port numbers

| System | Multicast Group | Product | ITCH Server-side IP Address | ITCH Server-side Port number |
|---|---|---|---|---|
| System 1 | MCG001 | Nikkei 225 Options, Nikkei 225 mini Options | 10.18.18.1 | 21814 |
| | MCG002 | Nikkei 225 mini | 10.18.18.3 | 21815 |
| | MCG003 | Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei 400 Futures | 10.18.18.5 | 21816 |
| | MCG004 | Other Index Futures/Options | 10.18.18.5 | 21826 |
| | MCG005 | Single Stock Options | 10.18.18.7 | 21817 |
| | MCG006 | JGB Futures, JGB Futures Options, Interest Rate Futures | 10.18.18.7 | 21827 |
| | MCG007 | OSE Commodity Futures (Precious Metals, Agriculture, Rubber, Gold Futures Options, etc.) | 10.18.18.7 | 21837 |
| | MCG008 | TOCOM Commodity Futures (Crude Oil, Power, LNG) | 10.18.18.9 | 21818 |
| System 2 | MCG001 | Nikkei 225 Options, Nikkei 225 mini Options | 10.18.19.2 | 21814 |
| | MCG002 | Nikkei 225 mini | 10.18.19.4 | 21815 |
| | MCG003 | Nikkei 225 Futures, Nikkei micro Futures, TOPIX Futures/Options, JPX-Nikkei400 Futures | 10.18.19.6 | 21816 |
| | MCG004 | Other Index Futures/Options | 10.18.19.6 | 21826 |
| | MCG005 | Single Stock Options | 10.18.19.8 | 21817 |
| | MCG006 | JGB Futures, JGB Futures Options, Interest Rate Futures | 10.18.19.8 | 21827 |
| | MCG007 | OSE Commodity Futures (Precious Metals, Agriculture, Rubber, Gold Futures Options, etc.) | 10.18.19.8 | 21837 |
| | MCG008 | TOCOM Commodity Futures (Crude Oil, Power, LNG) | 10.18.19.10 | 21818 |

(2) For production operation after a disaster recovery

Same as (1) For production operation

(3) For testing on market holidays

Same as (1) For production operation

### 4.4.3 User IDs and passwords for ITCH users

The user IDs and passwords required for ITCH users to log in to the J-GATE are specified as below.

Table_4.4.3 User IDs and Passwords

| Multicast Group | Product | User ID for Production Envr / DR / Holiday Production Test | Password for Production Envr / DR / Holiday Production Test |
|---|---|---|---|
| MCG001 | Nikkei 225 Options, Nikkei 225 mini Options | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | kc794ek4 |
| MCG002 | Nikkei 225 mini | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | tg494ym9 |
| MCG003 | Nikkei 225 Futures, Nikkei 225 micro Futures, TOPIX Futures/Options, JPX-Nikkei 400 Futures | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | wt649tn7 |
| MCG004 | Other Index Futures/Options | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | yu466we6 |
| MCG005 | Single Stock Options | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | jn936vk9 |
| MCG006 | JGB Futures, JGB Futures Options, Interest Rate Futures | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | ea763xr9 |
| MCG007 | OSE Commodity Futures (Precious Metals, Agriculture, Rubber, Gold Futures Options, etc.) | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | xi274cn7 |
| MCG008 | TOCOM Commodity Futures (Crude Oil, Power, LNG) | GLMP1、GLMP2、GLMP3、GLMP4、GLMP5、GLMP6、GLMP7、GLMP8、GLMP9、GLMP10 | gh318lq4 |

### 4.4.4 Data transmission procedure

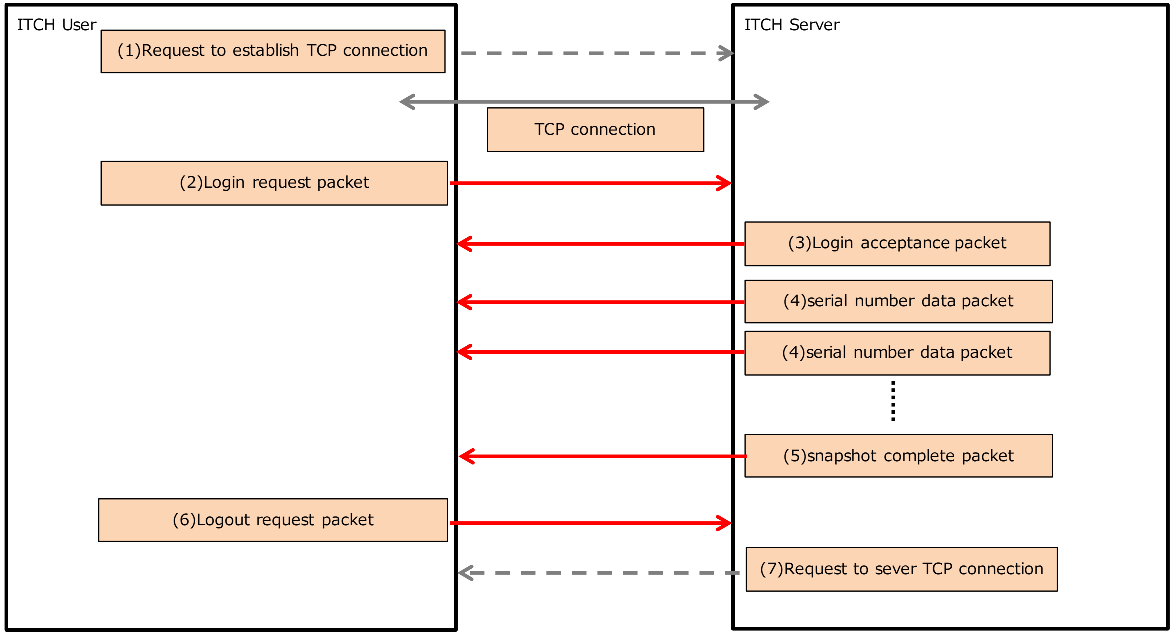
(1) Snapshot data procedure

The following table shows the procedures in which the ITCH user obtains snapshot data.

Table_4.4.4　Snapshot information process flow

| No. | Description |
|---|---|
| (1) | A TCP connection is established between the ITCH user and the ITCH server. (TCP connection establishment request is sent by the ITCH user.) |
| (2) | The ITCH user sends a login request packet. |
| (3) | If the login data is valid, the ITCH server sends a login acceptance packet. |
| (4) | The ITCH server sends data packets with continuous sequence numbers containing snapshot data. |
| (5) | The ITCH server sends a snapshot completion packet. |
| (6) | The ITCH user sends a logout request packet. |
| (7) | The TCP connection between the ITCH user and the ITCH server is terminated. (A TCP disconnection request is sent by the ITCH server.) |

Image_4.4.4-1　Snapshot information process flow



(2) Processing when the login data is invalid

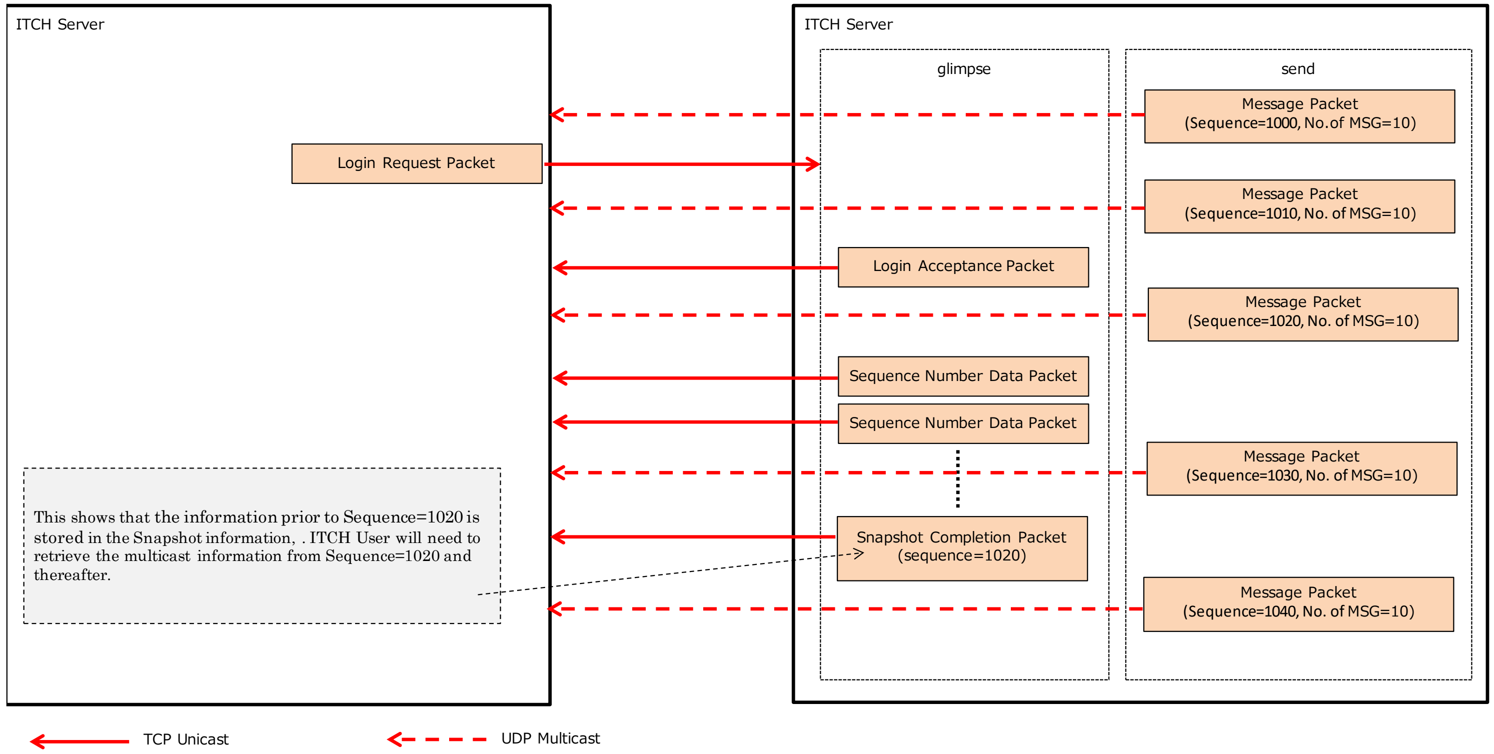
- When a login request packet is received from an ITCH user, the ITCH server checks the user name, password, and IP address of the ITCH user. If any of these are different from the pre-registered value, the ITCH server sends a login rejection packet and then terminates the TCP connection with the ITCH user.
- If a login request packet is not received from an ITCH user within 30 seconds of TCP connection establishment, the ITCH server terminates the TCP connection with the ITCH user.

(3) Receiving multicast data while receiving snapshot data

While data packets with continuous sequence numbers (snapshot data) are being sent, newly available data may sometimes be sent as multicast data, mixed in with the snapshot data. Therefore, when receiving data, the ITCH user needs to take sequencing with the snapshot data into consideration, and respond accordingly, such as by storing the multicast data in a temporary area.

Since data packets with continuous sequence numbers (snapshot data) sent from the ITCH server contain the most recent data at any given point in time, they do not incorporate data that became available after that point. As shown below, the message sequence numbers for newly available multicast data are stored in a snapshot completion packet, which is sent as soon as the ITCH server has finished sending data packets with continuous sequence numbers (snapshot data). Based on this sequence number, the ITCH user needs to capture or discard the multicast data.

Image_4.4.4-2　Multicast information receipt during snapshot information receipt process flow




(4) Heartbeat packet

When a login request packet is received normally from an ITCH user, the ITCH server and ITCH user then continue sending heartbeat packets to each other until the TCP connection is terminated.

If no packet is sent for 0.05 seconds or longer to the ITCH user, the ITCH server sends a heartbeat packet. In other words, there is always at least one packet sent per 0.05 seconds.

Likewise, if no packet is sent for 1 second or longer to the ITCH server, the ITCH user sends a heartbeat packet, so that at least one packet per second is always sent. If no packet is received from the ITCH user for 30 seconds, the ITCH server assumes that an error has occurred and terminates the TCP connection with the ITCH user.

### 4.4.5 Content of the packets that are transmitted

(1) Types of packets that are transmitted

The following table shows the types of packets that are transmitted:

Table_4.4.5-1　Transmitted packet

| No. | Packet type | Transmission source | | Transmission timing | Remarks |
|---|---|---|---|---|---|
| | | ITCH user | ITCH server | | |
| 1 | Login request packet | ○ | | When an ITCH user logs in to the ITCH server | |
| 2 | Login acceptance packet | | ○ | When a login request is received from an ITCH user | |
| 3 | Login rejection packet | | ○ | When a login request from an ITCH user is rejected | |
| 4 | Logout request packet | ○ | | When an ITCH user logs out of the ITCH server | |
| 5 | Data packet with continuous sequence number | | ○ | When the requested snapshot data is sent | |
| 6 | Snapshot transmission completion packet | | ○ | When transmission of snapshot data (in data packets with continuous sequence numbers) is completed | |
| 7 | User heartbeat packet | ○ | | When no packet is sent from a logged-in ITCH user to the ITCH server for 1 second or longer | |
| 8 | Server heartbeat packet | | ○ | When no packet is sent from the ITCH server to a logged-in ITCH user for 0.05 seconds or longer | |

(2) Settings

(a) Packets sent by the ITCH server

[1] Login acceptance packet

This packet is sent when a valid login request is received from an ITCH user.

Table_4.4.5-2　Login acceptance packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “A”: Login acceptance packet | |
| 3 | Session | 3 | 10 | Alpha | Same as *Session* in the packet header. | |
| 4 | Sequence Number | 13 | 20 | Alpha | The first sequence number (in ASCII format) of the snapshot data (data packets with continuous sequence numbers) that is sent (Left-aligned and filled to the right with spaces) | |

[2] Login rejection packet

This packet is sent when an invalid login request is received from an ITCH user. After sending this packet, the ITCH server terminates the TCP connection.

Table_4.4.5-3　Login rejection packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “J”: Login rejection packet | |
| 3 | Reject Reason Code | 3 | 1 | Alpha | Login rejection reason code<br>“A”: Invalid username or password<br>“S”: The requested session or market data group is invalid or not available. | |

[3] Data packets with continuous sequence numbers

After sending a login acceptance packet, the ITCH server stores the snapshot data in data packets with continuous sequence numbers and sends them.

Table_4.4.5-4　Data packets with continuous sequence numbers

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “S”: Data packets with continuous sequence numbers | |
| 3 | Message | 3 | Variable | Alpha | Message is stored in the same format as that used for message data sent as multicast data. Additionally, each message data item is preceded by Seconds tag (T tag).<br>*Refer to *Chapter 6* for details. | |

[4] Snapshot completion packet

The ITCH server sends this packet after completing the transmission of snapshot data, which uses data packets with continuous sequence numbers.

Table_4.4.5-5　Snapshot completion packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “S”: Data packets with continuous sequence numbers | |
| 3 | Message Type | 3 | 1 | Alpha | “G”: Snapshot completion packet | |
| 4 | Sequence | 4 | 20 | Alpha | The sequence number of the multicast data to be received just after snapshot data is received | |

[5] Server heartbeat packet

This packet is sent at 0.05-second intervals to check on communication when no packet has been sent from the server.

Table_4.4.5-6 Server heartbeat packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “H”: Server heartbeat packet | |

(b) Packets sent by the ITCH user

[1] Login request packet

The ITCH user sends this request packet to log in to the ITCH server.

Table_4.4.5-7 Login request packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “L”: Login request packet | |
| 3 | Username | 3 | 6 | Alpha | Username<br>(Username assigned to each ITCH user by the Osaka Exchange) | |
| 4 | Password | 9 | 10 | Alpha | Password<br>(Password assigned to each ITCH user by the Osaka Exchange) | |
| 5 | Requested Session | 19 | 10 | Alpha | Same as *Session* in the packet header.<br>If filled with spaces, the current active session is assumed. | |
| 6 | Requested Sequence Number | 29 | 20 | Alpha | Always set to 1. | |

[2] User heartbeat packet

This packet is sent at 1-second intervals to check on communication when no packet has been sent from the ITCH user.

Table_4.4.5-8 User heartbeat packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “R”: User heartbeat packet | |

[3] Logout request packet

This request packet is sent by the ITCH user to log out of the ITCH server.

Table_4.4.5-9 Logout request packet

| No. | Item name | Position | Number of bytes | Attribute | Description | Remarks |
|---|---|---|---|---|---|---|
| 1 | Packet Length | 0 | 2 | Numeric | Number of bytes in this packet excluding this item. | |
| 2 | Packet Type | 2 | 1 | Alpha | “O”: Logout request packet | |

## 4.4.6 Other

(1) Snapshot data can be obtained only while the ITCH server is online (between 06:39 am on a business day and 06:20 am on the following day)

(2) Only the snapshot data of the current day can be obtained.

(3) If the ITCH server reaches the end of its online operation (around 06:20 am), while an ITCH user is still logged in, the TCP connection between the ITCH user and the ITCH server is terminated.

(4) The usage of snapshot data request of each participant will be checked on regular basis.

# 5. During Disaster Recovery

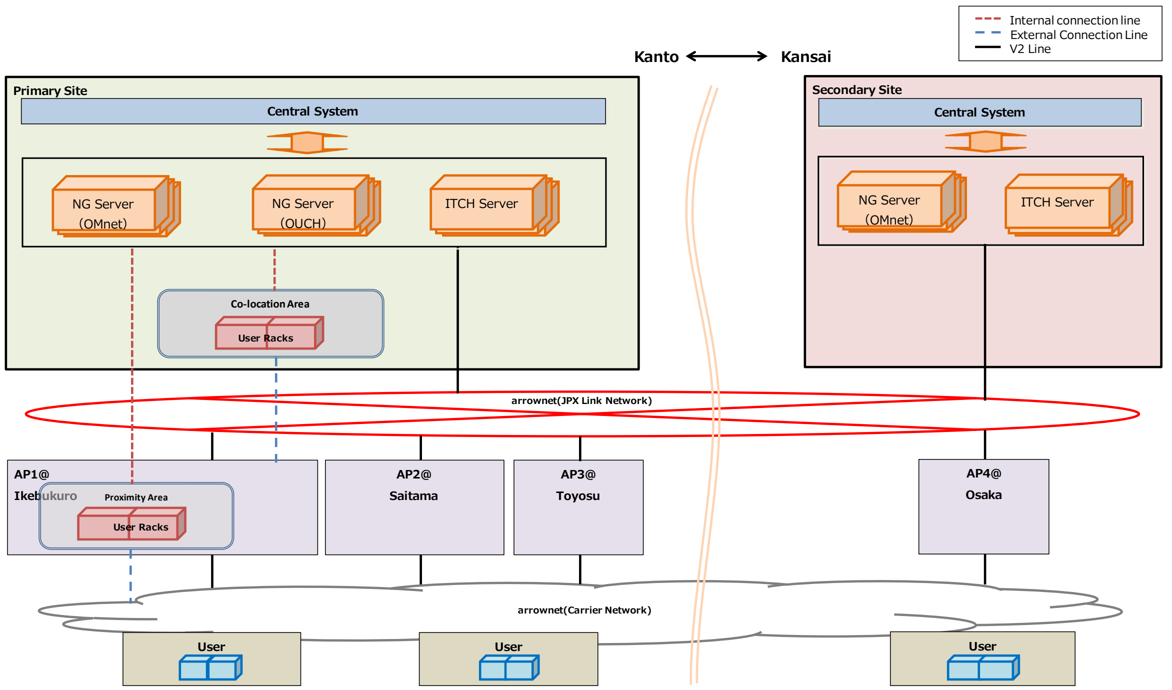
## 5.1 During Disaster Recovery

A backup server is provided to allow trading operations to continue, in the event that the primary site is struck by a localized or wide-area disaster and cannot continue trading operations. If a disaster occurs at the primary site and the business continuity plan (BCP) is activated, after system switchover to the backup site is completed[1], ITCH users can connect to the backup site's ITCH gateway server using the same settings that were used during normal operation (before the disaster).

Note that after a disaster occurs at the primary site, data for the day of the disaster cannot be resent using the *rewind* or *glimpse* option.

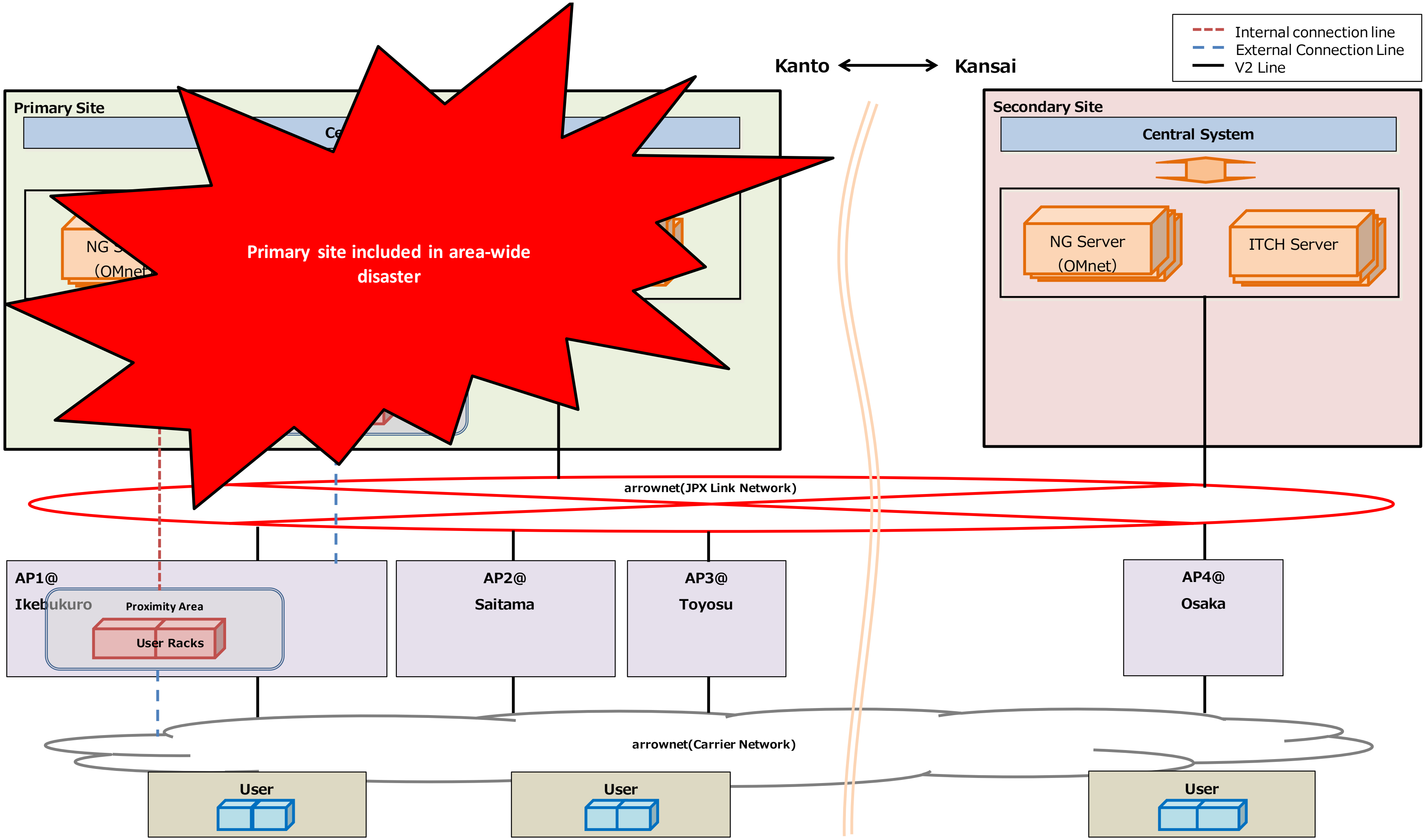
The following figure provides an overview of switchover to a backup site.

Image_5-1 During normal operation




[1] When connection to all external systems has been completed at the backup site

Image_5-2　When a disaster occurs at the primary site



## 5.2 Policy on Responses to System Failures

### 5.2.1 Basic Policy

Response measures will be taken in accordance with “3.3 Policy on responses to system failures” in “J-GATE3.0 (Derivatives trading system) Connectivity Manual (Commonalities part)”.

### 5.2.2 Measures for a System Failure where the Exchange cannot foresee the impacts of the failure

Response measures will be taken in accordance with “3.3.2.1 Response Measures for Non-acceptance of Orders, Trading Suspension, etc.” in J-GATE3.0 (Derivatives trading system) Connectivity Manual (Commonalities part)”.

All ITCH functionalities may become unavailable, depending on failure situations.

### 5.2.3 Recovery by System Reboot

Response measures will be taken in accordance with “3.3.2.4 Recovery by system reboot” in “J-GATE3.0 (Derivatives trading system) Connectivity Manual (Commonalities part)”.

Trading status information tag is disseminated every time TSS is changed following a system reboot. In addition, the data disseminated before the reboot is no longer available for rewind or glimpse after the reboot.

### 5.2.4 Recovery for ITCH Failure

When recovery from a single-side system failure is possible, priority will be given to promptly returning to the state of dual systems reception, and recovery may take place during auction hours. It is assumed that the Exchange will make an announcement in advance regarding expected detail of recovery and time frame. When recovering from a prolonged single-side system failure, business operations should be resumed using message which is received normally via the other side system or by using Glimpse. At this time, using of Rewind should be avoided as much as possible as it may cause repeated message requests which may place an excessive load on the ITCH server.

# 6. Data Format
## 6.1 Message types

There are four ITCH data types, as described in the following table.

| Message type | Size | Description |
|---|---|---|
| Numeric | 1, 2 , 4 or 8 bytes | Numeric value encoded in big-endian binary.<br>Note: SoupBinTCP and Mold UDP, which are transport layer protocols, use big-endian. |
| Alpha | Variable length | Set left-aligned and filled to the right with spaces |
| Price | 4 bytes | An integer value is set. The decimal position is set in the series information.<br>Note: If the 31-bit minimum value (-2147483648) is set in this field, this value is treated as meaning No_Value (no value).<br>No_Value is also set for new market orders. |
| Day | 4 bytes | An integer value is set in the *YYYYMMDD* format. |

## 6.2 Timestamps

ITCH disseminates timestamps indicating the times when issuance, cancellation, execution, etc. of orders are performed by the Matching Engine.
Timestamps are also provided when series information and session information are disseminated.
To more efficiently utilize the line capacity, ITCH divides timestamps into two segments, as explained below.

| Timestamp segment | Message type | Remarks |
| --- | --- | --- |
| Seconds | Stand-alone message (disseminated individually) | UNIX time (The time elapsed since January 1, 1970 00:00:00 GMT is output.)<br>This message is added in second units when individual messages are disseminated. |
| Nanoseconds | Embedded and disseminated in individual messages | Sets the timestamp that is closest to the creation time of the last of the disseminated messages.<br>A specific example follows:<br>525530183 |

## 6.3 Detailed specifications for data items in each tag (general)

### 6.3.1 Seconds tag (Tag ID: T)

(1) Tag content

Provides data on the seconds when each message is disseminated. If multiple messages are disseminated in the same second, this tag is disseminated only once before the first message.

(2) Tag output timing

This tag is output when any message is disseminated from ITCH.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | "T" is set, indicating the second tag. | T | |
| 2 | Second | 1 | 4 | Numeric | UNIX time (The time elapsed since January 1, 1970 00:00:00 GMT is output.) | 1438807866 | |

### 6.3.2 Series information basic tag (Tag ID: R)

(1) Tag content

Provides detailed data about the series that can be traded on a given business day (including series for which trading is currently suspended).

(2) Tag output timing

This tag is output after a certain amount of time has elapsed following the start of the online system operation.

Will be provided when Series Information in J-GATE is updated (It also may be provided when information which is not output by R-tag is updated.)

(3) Provided Item Details

<table>
<tr><th>No.</th><th>Item name</th><th>Position</th><th>Number of bytes</th><th>Attribute</th><th>Description</th><th>Setting example</th><th>Remarks</th></tr>
<tr><td>1</td><td>Message Type</td><td>0</td><td>1</td><td>Alpha</td><td>“R” is set, indicating the series data basic tag.</td><td>R</td><td></td></tr>
<tr><td>2</td><td>Timestamp – Nanoseconds</td><td>1</td><td>4</td><td>Numeric</td><td>A timestamp is set in nanoseconds.</td><td>826482921</td><td></td></tr>
<tr><td>3</td><td>Order book ID</td><td>5</td><td>4</td><td>Numeric</td><td>A series ID is set.</td><td>184615012</td><td>*1, *2</td></tr>
<tr><td>4</td><td>Symbol</td><td>9</td><td>32</td><td>Alpha</td><td>A series name is set.</td><td>PUT_225_220609_19000</td><td></td></tr>
<tr><td>5</td><td>Long Name</td><td>41</td><td>32</td><td>Alpha</td><td>A 9-digit code is set.</td><td>137069018</td><td></td></tr>
<tr><td>6</td><td>Reserved</td><td>73</td><td>12</td><td>Alpha</td><td>Reserved item</td><td></td><td></td></tr>
<tr><td>7</td><td>Financial Product</td><td>85</td><td>1</td><td>Numeric</td><td>A product type is set.<br><table><tr><th>Code</th><th>Explanation of settings</th></tr><tr><td>1</td><td>Option</td></tr><tr><td>3</td><td>Futures</td></tr><tr><td>11</td><td>Combination Series</td></tr></table></td><td>1</td><td></td></tr>
<tr><td>8</td><td>Trading Currency</td><td>86</td><td>3</td><td>Alpha</td><td>A trading currency is set. Fixed at JPY.</td><td>JPY</td><td></td></tr>
<tr><td>9</td><td>Number of decimals in Price</td><td>89</td><td>2</td><td>Numeric</td><td>The decimal value for a trading instrument is set.</td><td>4</td><td></td></tr>
</table>

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 10 | Number of decimals in Nominal Value | 91 | 2 | Numeric | Fixed at 0 | 0 | |
| 11 | Odd Lot Size | 93 | 4 | Numeric | Fixed at 0 | 0 | |
| 12 | Round Lot Size | 97 | 4 | Numeric | Fixed at 1 | 1 | |
| 13 | Block Lot Size | 101 | 4 | Numeric | Fixed at 0 | 0 | |
| 14 | Nominal Value | 105 | 8 | Numeric | Fixed at 0 | 0 | |
| 15 | Number of Legs | 113 | 1 | Numeric | 0 is set for Normal Series.<br>Number of legs are set for combination series | 0 | |
| 16 | Underlying Order book ID | 114 | 4 | Numeric | The underlying code is set.<br>0 is set for combination series. | 100 | |
| 17 | Strike Price | 118 | 4 | Price | Strike Price is set. If the case is futures, 0 is set.<br>Sequential number of when it is created is set for TMC. | 19000 | |
| 18 | Expiration Date | 122 | 4 | Day | Expiration date is set. | 20220610 | |
| 19 | Number of decimals in Strike Price | 126 | 2 | Numeric | Number of decimals in strike price is set. | 0 | |
| 20 | Put or Call | 128 | 1 | Numeric | Distinction of Put or Call is set. If the case is futures or combination series, 0 is set.<br>1: Call 2: Put | 2 | |

*1: Might overlap with a past series that expired.

*2: Since the series ID is set in API DQ124 and DQ126, various types of data can be linked by using the series ID.

### 6.3.3 Series information basic tag: Combination trading instruments (Tag ID: M)

(1) Tag content

Provides detailed data about the legs of combination trading instruments that can be traded on a given trading day (including series for which trading is currently suspended).

For each leg constituting the combination trading instrument, this tag is output in separate records to provide data on distinction of buy leg and sell leg and the ratio.

Combination trading instrument of the leg is judged using the Order book ID in R tag of that Combination series and Combination Order book ID.

(2) Tag output timing

This tag is output after a certain amount of time has elapsed following the start of the online system operation.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | “M” is set, indicating the Series information basic tag: Combination trading instruments. | M | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Combination Order book ID | 5 | 4 | Numeric | A series ID of Combination trading instrument is set. | 131502 | *1 |
| 4 | Leg Order book ID | 9 | 4 | Numeric | Series ID of the leg is set. | 328110 | *2 |
| 5 | Leg Side | 13 | 1 | Alpha | The buy or sell side of Leg is set.<br>B: Buy if on buy side (sell if on sell side)<br>C: Sell if on buy side (buy if on sell side) | B | |
| 6 | Leg Ratio | 14 | 4 | Numeric | Ratio of leg is set. | 2 | |

*1: Same code is set for each leg constituting the same combination trading instrument.

*2: Since the series ID is set in Series information basic tag (R-Tag), various types of data can be linked by using the series ID.

### 6.3.4 Tick size data tag (Tag ID: L)

(1) Tag content

Provides the tick size of a trading instrument.

(2) Tag output timing

This tag is output after a certain amount of time has elapsed following the start of online system operation.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | "L" is set, indicating the tick size data tag. | L | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order book ID | 5 | 4 | Numeric | A series ID is set. | 184615012 | |
| 4 | Tick Size | 9 | 8 | Numeric ~~Price~~ | A tick size that is between the values in No.5 and No.6 is set. | 10000 | *1 |
| 5 | Price From | 17 | 4 | Price | The smallest value within the price range to which the applicable order price applies is set. | 10000 | *2 |
| 6 | Price to | 21 | 4 | Price | The largest value within the price range to which the applicable order price applies is set. | 999999 | *2 |

*1: If the order price varies depending on the price range, multiple records are output. See next page for the specifics.

*2: Values will be provided with decimals considered.

| Message Type | Timestamp – Nanoseconds | Order book ID | Tick Size | Price From | Price to |
|---|---|---|---|---|---|
| L | 525530183 | 196888 | 100 | 100 | 999999900 |
| L | 525530183 | 525594 | 1 | 1 | 499 |
| L | 525530183 | 525594 | 5 | 500 | 9999 |
| L | 525530183 | 525594 | 10 | 10000 | 29999 |
| L | 525530183 | 525594 | 50 | 30000 | 299999 |
| L | 525530183 | 525594 | 250 | 300000 | 499999 |
| L | 525530183 | 525594 | 500 | 500000 | 999999 |
| L | 525530183 | 525594 | 5000 | 1000000 | 9999999 |
| L | 525530183 | 525594 | 50000 | 10000000 | 999950000 |

### 6.3.5 System event data tag (Tag ID: S)

(1) Tag content

Provides system event update data.

(2) Tag output timing

This tag is output when a system event is updated.

Refer to 4.2.6 (4) for details.

(3) Details about the content of the items provided by the tag

<table>
<tr><th>No.</th><th>Item name</th><th>Position</th><th>Number of bytes</th><th>Attribute</th><th>Description</th><th>Setting example</th><th>Remarks</th></tr>
<tr><td>1</td><td>Message Type</td><td>0</td><td>1</td><td>Alpha</td><td>“S” is set, indicating the system event data tag.</td><td>S</td><td></td></tr>
<tr><td>2</td><td>Timestamp – Nanoseconds</td><td>1</td><td>4</td><td>Numeric</td><td>A timestamp is set in nanoseconds.</td><td>826482921</td><td></td></tr>
<tr><td>3</td><td>Event Code</td><td>5</td><td>1</td><td>Alpha</td><td>An event code is set.<table><tr><th>Code</th><th>Explanation of settings</th></tr><tr><td>O</td><td>Start of message transmission. All business days start with the transmission of this message.</td></tr><tr><td>C</td><td>End of message transmission. All business trading days end with the transmission of this message.</td></tr></table></td><td>O</td><td>*1</td></tr>
</table>

*1: For this tag, only two codes are sent in one day, i.e., code O for the start and code C for the end of a business day.

### 6.3.6 Trading status data tag (Tag ID: O)

(1) Tag content

Provides the trading status data of a trading instrument.

(2) Tag output timing

This tag is output for each series when a trading status is updated.

In the case where consecutive DCB occurs, status for ZARABA will be provided once in between transition from DCB to another DCB.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | "O" is set, indicating the trading status data tag. | O | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order book ID | 5 | 4 | Numeric | A series ID is set. | 6226220 | |
| 4 | State Name | 9 | 20 | Alpha | Status data is set. | M_PRE_OptionsEN_NO_J-NET | |

## 6.4 Detailed specifications for data items in each tag (orders)

This section describes the data formats of messages related to orders and executions.

The order number added to each message is unique for each order book and buy/sell side, but it is not unique in all other cases. Therefore, when there are multiple order books or when the buy/sell sides are different, it is possible that the same order number might be set in cases where an order is corrected or deleted.

### 6.4.1 New order tag (Tag ID: A)

(1) Tag content

Provides quote data related to a new order.

(2) Tag output timing

This tag is output when a new order is issued.

To correct an order, a new order tag with the correct order is output after the incorrect order is cancelled with the Deleted Order tag.

Every morning, new order tag for reloaded GTD/GTC orders are output after Series information basic tag is provided.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | “A” is set, indicating the new order tag. | A | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order ID | 5 | 8 | Numeric | An order number is set. (The order number is unique for each order book and sell/buy side.) | 7400681928971580000 | |
| 4 | Order book ID | 13 | 4 | Numeric | A series ID is set. | 6226220 | |
| 5 | Side | 17 | 1 | Alpha | Sell or Buy is set.<br>B: Buy; S: Sell | B | |
| 6 | Order Book Position | 18 | 4 | Numeric | Order priority is set. | 3 | |
| 7 | Quantity | 22 | 8 | Numeric | A new order quantity is set. | 10 | |
| 8 | Price | 30 | 4 | Price | A new order price is set. | 1980000 | * |
| 9 | Order Attributes | 34 | 2 | Numeric | Item not used. | 0 | |
| 10 | Lot Type | 36 | 1 | Numeric | Fixed at 2 | 2 | |

*: Values in Price will be provided with decimals considered. Also, -2147483648 is set to market orders during pre-trading period.

### 6.4.2 Execution Notice tag (Tag ID: E)

(1) Tag content

This tag provides an execution notice.

The Order ID of the order which is already in the order book is set, used to get the order information.

E.g., If there is a sell order on the order book, and it is executed by a buy order, the Order ID of the sell order is set.

(2) Tag output timing

This tag is output whenever an order is partially or fully executed.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | “E” is set, indicating the execution notice tag. | E | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order ID | 5 | 8 | Numeric | An order number is set.<br>(The order number is unique for each order book and buy/sell side.) | 7409390335941466259 | |
| 4 | Order book ID | 13 | 4 | Numeric | A series ID is set. | 6226220 | |
| 5 | Side | 17 | 1 | Alpha | Buy or Sell is set.<br>B: Buy; S: Sell | B | *1 |
| 6 | Executed Quantity | 18 | 8 | Numeric | An executed quantity is set. | 15 | |

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 7 | Match ID | 26 | 8 | Numeric | An ExecutionEventNumber is set. | 656706413728366001 | *2 |
| 8 | Combo Group ID | 34 | 4 | Numeric | In the case of a combination series execution, a combo group ID is set. Otherwise, 0 is set. | 10 | |
| 9 | Reserved | 38 | 7 | Alpha | Reserved item. Fixed to space. | (space) | |
| 10 | Reserved | 45 | 7 | Alpha | Reserved item. Fixed to space. | (space) | |

*1: During Zaraba, the side of the order which is already in the order book is set (If there is a sell order on the order book, and it is executed by a buy order, “S” is set).

*2: The Execution notification number is the same as the value set in BD70.

### 6.4.3 Execution notice tag with trade information (Tag ID: C)

(1) Tag content

Provides an execution notice with trade information.

During Zaraba, the side of the order which is already in the order book is set. (If there is a sell order in the order book, and it is executed by a buy order, “S” is set).

(2) Tag output timing

This tag is output when combination series executes with another combination series, and at executions during auction. It is also output when execution occurs at a price different than initially ordered.

-If a combination series executes with a combination series, an execution notice tag with trade information (C-tag) will be disseminated.

-If multiple orders execute in auction, C-tag will be disseminated for all executed buy trades and sell trades regardless of which order has been in the order book first.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | “C” is set, indicating the execution notice tag with trade information. | C | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order ID | 5 | 8 | Numeric | An order number is set.<br>(The order number is unique for each order book and sell/buy side.) | 7409390335941466259 | |
| 4 | Order book ID | 13 | 4 | Numeric | A series ID is set. | 6226220 | |

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 5 | Side | 17 | 1 | Alpha | Buy or Sell is set.<br>B: Buy; S: Sell | B | *1 |
| 6 | Executed Quantity | 18 | 8 | Numeric | An executed quantity is set. | 15 | |
| 7 | Match ID | 26 | 8 | Numeric | An ExecutionEventNumber is set. | 656706413728366001 | |
| 8 | Combo Group ID | 34 | 4 | Numeric | In the case of a combination series execution, a combo group ID is also set. | 10 | *2 |
| 9 | Reserved | 38 | 7 | Alpha | Reserved item | (space) | |
| 10 | Reserved | 45 | 7 | Alpha | Reserved item | (space) | |
| 11 | Trade Price | 52 | 4 | Price | An execution price is set. | 1950000 | |
| 12 | Occurred at Cross | 56 | 1 | Alpha | Identification flag for auction<br><table><tr><th>Code</th><th>Explanation of settings</th></tr><tr><td>Y</td><td>Execution through auction</td></tr><tr><td>N</td><td>Execution through Zaraba</td></tr></table> | N | |
| 13 | Printable | 57 | 1 | Alpha | Item not used. | N | |

*1: During Zaraba, the side of the order which is already in the order book is set

### 6.4.4 Deleted order tag (Tag ID: D)

(1) Tag content

This tag is provided when an order is deleted and does not exist in the order book.

(2) Tag output timing

This tag is normally output when an order has been deleted.

To correct an order, a new order tag with the correct order is output after the incorrect order is cancelled with the Deleted Order tag.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | “D” is set, indicating the deleted order tag. | D | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order ID | 5 | 8 | Numeric | An order number is set.<br>(The order number is unique for each order book and sell/buy side.) | 7409390335941466259 | |
| 4 | Order book ID | 13 | 4 | Numeric | A series ID is set. | 6226220 | |
| 5 | Side | 17 | 1 | Alpha | Buy or Sell is set.<br>B: Buy; S: Sell | B | |

### 6.4.5 EP tag (Tag ID: Z)

(1) Tag content

Provides the equilibrium price (EP) for the order reception hours.

(2) Tag output timing

This is output whenever tag contents (including spare items) are updated during order reception hours, such as during order reception, during DCB, and during trade suspension.

(3) Details about the content of the items provided by the tag

| No. | Item name | Position | Number of bytes | Attribute | Description | Setting example | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Message Type | 0 | 1 | Alpha | "Z" is set, indicating the EP tag. | Z | |
| 2 | Timestamp – Nanoseconds | 1 | 4 | Numeric | A timestamp is set in nanoseconds. | 826482921 | |
| 3 | Order book ID | 5 | 4 | Numeric | A series ID is set. | 6226220 | |
| 4 | Available Bid Quantity at Equilibrium Price | 9 | 8 | Numeric | Total bid order quantity at the EP is set. | 300 | |
| 5 | Available Ask Quantity at Equilibrium Price | 17 | 8 | Numeric | Total ask order quantity at the EP is set. | 250 | |
| 6 | Equilibrium Price | 25 | 4 | Price | The EP is set. | 19800 | * |
| 7 | Reserved | 29 | 4 | | Reserved item.<br>If any value is set, ignore it. | | |
| 8 | Reserved | 33 | 4 | | Reserved item.<br>If any value is set, ignore it. | | |
| 9 | Reserved | 37 | 8 | | Reserved item.<br>If any value is set, ignore it. | 0 | |
| 10 | Reserved | 45 | 8 | | Reserved item.<br>If any value is set, ignore it. | 0 | |

*If EP does not exist, -2147483648 is set.

# 7. ITCH Connectivity Compliance and Prohibitions

## 7.1 Infinite Loop Process is prohibited (building a business system that does not generate infinite loops)

It is possible to end up with a retry loop when an error occurs in applications on the user side. In this case, the number of retries must be limited, and a mechanism must be implemented to automatically handle any endless loops which repeat the same content communication using the API.

The following business process can be considered as a specific example of creating the retry process.

—Issue a Rewind

—Issue a GLIMPSE

## 7.2 Rewind and GLIMPSE Usage in ITCH

Since ITCH Rewind is used for recovery in the event of information loss or participant system failure, in principle, it should not be used for other purposes. In particular, daily acquisition of all messages after the end of trading is prohibited.

GLIMPSE is used to get the price indication information at the time of failure recovery. Therefore, it is imperative that it not be used for other purposes. In order to obtain morning series information and information on carryover orders, please join the multicast group before starting online in the morning, as described in this manual (Glimpse usage is only allowed when the applicable information could not be acquired).

# 8. Change Log

## 8.1 Change Log to Version 1.1

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2020/11 | 1.2 Overview of the messages provided by the system | Added description below<br><br>We make every effort to prevent an incident in J-GATE. However, even if ITCH is down, we, in principle, continue trading as long as OMnet API is running to provide market data. (Except in cases where more than a certain ratio of trading participants cannot participate in trading.) | |
| 2 | 2020/11 | 6.4.3 Execution notice tag with trade information (Tag ID: C) | Modified description on "(2) Tag output timing"<br><br>Before:<br>-If a combination series executes with a combination series, an execution notice tag with trade information (C-tag) and as many times as the number of legs of the combination series the price notification tag (P-tag) which notifies the execution price of the legs of the combination series will be disseminated.<br><br>After:<br>-If a combination series executes with a combination series, an execution notice tag with trade information (C-tag) will be disseminated. | |

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 3 | 2020/11 | 6.4.6 EP tag (Tag ID: Z) | Modified "Description" on 7-9 (Reserved)<br><br>Before: Fixed at 0<br>After: If any value is set, ignore it. | |
| 4 | 2020/11 | Image_2.1 J-GATE Overview<br>Image_5-1 During normal operation<br>Image_5-2 When a disaster occurs at the primary site | Added "AP4" to the image | |
| 5 | 2020/11 | All | Made minor corrections of words and phrases to make them more consistent | |
| 6 | 2020/11 | Reference_ITCH Case examples.xlsx | Added "Reference_ITCH Case examples.xlsx" | |

## 8.2 Change Log to Version 1.2

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2021/2 | Table_4.3.2 IP addresses and port numbers | Modified "ITCH Server-side Port number" | |
| 2 | 2021/2 | Table_4.4.2 IP addresses and port numbers | Modified "ITCH Server-side Port number" | |
| 3 | 2021/2 | Table_4.4.3 User IDs and passwords | User ID for production operation, disaster recovery, and testing on market holidays | |
| 4 | 2021/2 | 6.3.3 Series information basic tag: Combination trading instruments (Tag ID: M) | Deleted description below from “(2) Tag output timing”<br><br>It is provided when a flex series is generated. | |
| 5 | 2021/2 | 6.3.4 Tick size data tag (Tag ID: L) | Added description below to “(2) Tag output timing”<br><br>It is provided when a flex series is generated. | |
| 6 | 2021/2 | 6.3.6 Trading status data tag (Tag ID: O) | Added description below to “(2) Tag output timing”<br><br>It is provided when a flex series is generated. | |
| 7 | 2021/2 | 6.4.3 Execution notice tag with trade information (Tag ID: C) | Deleted description below<br><br>*2: Price notification tag and execution notice tag with trade information is to be matched by Combo Group ID and Match ID | |

## 8.3 **Change Log to Version 1.3**

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2021/4 | 4.2.5 Content of the packets that are transmitted<br>4.2.6 Other<br>4.3.5 Other<br>4.4.6 Other | Added description of time | |
| 2 | 2021/4 | 6.3 Detailed specifications for data items in each tag (general)<br>6.4 Detailed specifications for data items in each tag (orders) | Modified “Setting example” of each tag | |
| 3 | 2021/4 | 6.3.2 Series information basic tag (Tag ID: R)<br>6.3.4 Tick size data tag (Tag ID: L)<br>6.3.6 Trading status data tag (Tag ID: O) | Deleted description below<br><br>Will be provided when flex series is generated. | |

## 8.4 Change Log to Version 1.4

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2021/6 | 4.2.2 Maximum number of messages per second and maximum bandwidth for multicast groups | Added the following description:<br><br>It may momentarily exceed the assumed values depending on market conditions. If disseminated messages exceed the assumed bandwidth and a packet loss is detected, send a rewind request to recover the missed information. | |
| 2 | 2021/6 | 5.2 Policy on responses to system failures | Added the chapter. | |

## 8.5 Change Log to Version 1.5

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
| --- | --- | --- | --- | --- |
| 1 | 2021/12 | Table_4.2.2-1 Assumed ITCH maximum number per second and the assumed required bandwidth<br>Table_4.2.2-2 Assumed required bandwidth when series information R tag and such is disseminated<br>Table_4.2.3-1 IP addresses and port numbers | Added LNG to P5 (MCG8) due to the launch of LNG (Platts JKM) Futures. | |
| 2 | 2021/12 | 4.3.2 IP addresses and port numbers<br>4.4.2 IP addresses and port numbers<br>4.4.3 User IDs and passwords for ITCH users | Added LNG to P5 (MCG8) due to the launch of LNG (Platts JKM) Futures. | |

## 8.6 Change Log to Version 1.6

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2021/12 | 1.4 Handling of this manual on Holiday Trading day | Added the chapter. | |
| 2 | 2021/12 | Table_4.2.5-2　Settings | Added descriptions about system reboot. | |
| 3 | 2021/12 | 5.2.3 Recovery by system reboot | Added the chapter. | |
| 4 | 2021/12 | J-GATE3.0 ITCH Connectivity Manual | Made some minor modifications. | |

## 8.7 Change Log to Version 1.7

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2022/4 | 4.3.5 Other | Changed description regarding arrownet provider from TSE to JPXI | |
| 2 | 2022/4 | - | From this version forth, only file with modification history will be provided via System Documents Site.<br>Your understanding and cooperation is highly appreciated. | |

## 8.8 Change Log to Version 1.8

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2022/6 | Image_2.2 Itch Server Example (Nikkei 225 Options, Nikkei 225 mini Options) | • Changed and added products due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 2 | 2022/6 | Table_4.2.2-1 Assumed ITCH maximum number per second and assumed required bandwidth | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 3 | 2022/6 | Table_4.2.2-2 Assumed required bandwidth when series Information R tag and such is disseminated | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 4 | 2022/6 | Table_4.2.3-1 IP address and port numbers | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 5 | 2022/6 | Table_4.3.2 IP addresses and port numbers | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 6 | 2022/6 | Table_4.4.2 IP addresses and port numbers | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |
| 7 | 2022/6 | Table_4.4.3 User IDs and passwords for ITCH users | • Changed and added products compositing MCG due to listing of Nikkei 225 micro Futures and Nikkei 225 mini Options (Q2 2023 release) | |

## 8.9 **Change Log to Version 1.9**

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2022/10 | Table_4.2.2-1 Assumed ITCH maximum number per second and the assumed required bandwidth | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |
| 2 | 2022/10 | Table_4.2.2-2 Assumed required bandwidth when series information R tag and such is disseminated | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |
| 3 | 2022/10 | Table_4.2.3-1 IP addresses and port numbers | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |
| 4 | 2022/10 | Table_4.3.2 IP addresses and port numbers | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |
| 5 | 2022/10 | Table_4.4.2 IP addresses and port numbers | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |
| 6 | 2022/10 | Table_4.4.3 User IDs and passwords | ・ Changed and added products compositing MCG due to listing of 3-Month TONA Futures (May 2023 Release) | |

## 8.10 Change Log to Version 2.0

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2024/12 | Table_4.4.3 User IDs and Passwords | ・ Revised misdescription regarding ITCH User IDs and Passwords | |
| 2 | 2024/12 | 5.2 Policy on Responses to System Failures | ・ Newly Published [5.2.4 Recovery for ITCH Failure] and added description regarding recovery timing when single-side system failure occurred | |

## 8.11 Change Log to Version 2.1

| No. | Modification Date | Modified Location (Chapter) | Modification Description | Remarks |
|---|---|---|---|---|
| 1 | 2025/4 | 6.3.4 Tick size data tag (Tag ID: L) | ・ Revised due to misdescription revision on J-GATE3.0 NASDAQ Manual [ITCH_ProtSpec_OSE_va70]. Revised misdescription regarding Tick Size attribute. | |